Write short notes on :

(a) Cylinder surface indexing.

(b) Trie tree indexing

(c) Linked file organization.

B.Tech. DEGREE EXAMINATION,
NOVEMBER/DECEMBER 2014.

Third Semester

Computer Science and Engineering

DATA STRUCTURES

Time : Three hours — Maximum : 75 marks

PART A — ($10 \times 2 = 20$ marks)

Answer ALL questions.

All questions carry equal marks.

1. Compare the order of growth $n!$ and 2^n.

2. What is multi-dimensional array?

3. Distinguish between Stack and Queue.

4. What is dequeue?

5. What are the applications of binary tree?

6. Define minimum spanning tree.

7. What is meant by Hash tables?

8. Differentiate internal and external sorting.

9. Define B+ tree.

10. What is the use of indexing technique?

PART B — (5 × 11 = 55 marks)

Answer ALL questions, ONE from each Unit.

All questions carry equal marks.

UNIT I

11. Explain the various algorithmic notations used in algorithm design.

Or

12. Write an algorithm for searching an element using binary search method. Give an example.

UNIT II

13. Define doubly linked list. Explain the insertion and deletion operation of double linked list with algorithm.

Or

14. Write the equivalent postfix expression for the following expressions using expression evaluation algorithm :

(a) $(A - B)/((D + E) * F)$ (6)

(b) $((A + B)/D) \uparrow ((E - F) * G$ (5)

UNIT III

15. Discuss the different methods of traversing a binary tree with algorithm.

Or

16. Explain the Kruskal's algorithm for computing the minimal spanning tree for weighted undirected graph.

UNIT IV

17. Compare the worst case and best case time complexity of various sorting techniques.

Or

18. Explain the collision resolution techniques in hashing.

UNIT V

19. What is B-Tree? Write an algorithm to insert an element into the B-Tree.

Or

UNIT IV

17. Explain the procedure to find the shortest path for the given graph. Develop and analyse your procedure.

Or

18. Describe the topological sort with an example.

UNIT V

19. Describe the different methods available to define the Hash function.

Or

20. (a) Distinguish between sequential and direct files. (5)

(b) List out the properties of an indexed sequential file and explain. (6)

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DATA STRUCTURES

Time : Three hours Maximum : 75 marks

PART A — ($10 \times 2 = 20$ marks)

Answer ALL questions.

All questions carry equal marks.

1. Define : Time complexity.

2. What is the average and worst case complexity of bubble sort?

3. Convert the following expression into reverse polish notation :

$(a-b)(c(d+e-f(g/h)))$.

4. Write the advantages of linked list.

5. What is the maximum number of nodes in a binary tree of depth d?

6. Consider the reserved words {if, while, loop, repeat, for}. Draw any two possible binary search tree.

7. What are the two ways to maintain the graph G in the memory of a computer?

8. When the graph is said to be strongly connected?

9. What is the use of symbol table?

10. What is latency time?

PART B — (5 × 11 = 55 marks)

Answer ALL questions, ONE question from each Unit.

All questions carry equal marks.

UNIT I

11. (a) Describe the criteria to be satisfied by an algorithm. (5)

(b) Describe the binary search procedure with an example. (6)

Or

12. Apply merge sort procedure for the following data :

(310, 285, 179, 652, 351, 423, 861, 254, 450, 520)

Also explain and develop an algorithm for the same.

UNIT II

13. (a) How the stack is useful to evaluate the expression? (5)

(b) How can you represent the queue as a linked list? Also write the procedure to add and delete an element from the linked queue. (6)

Or

14. (a) How can you perform polynomial addition using linked list? (6)

(b) Describe the different tasks of memory compaction. (5)

UNIT III

15. How can you represent the set as a tree? What are the operations that we can perform on these sets? How these operations are implemented?

Or

16. (a) Describe the insertion algorithm for B-trees. (5)

(b) Illustrate the deletion operation on B^{+}–tree and explain. (6)